Information Theory

An Introduction with a Tour of Surprises

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Abstract

These first lectures present, informally, a small set of results from information theory that contradict what most people expect before they see the proofs.

Every result stated here will be proved later in the course, in the proper order and with the proper hypotheses.

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1 Introduction

Information theory answers questions of the following kind.

- How compactly can a message be written down without losing anything? **(lossless compression)**
- How fast can a message be sent through a channel that corrupts it? **(reliable communication through noisy channels)**
- How much of a continuous signal survives when we record only finitely many numbers per second? **(discrete sampling of a continuous signal)**

Each of these questions has a sharp answer. In each case the answer is a single number attached to the source or to the channel, and in each case the answer is not the one that intuition suggests. That is the theme of these first lectures.

Information theory is the basis of most modern communication, storage and processing technology. A short list of areas where it is used:

- Efficient representation, transmission and retrieval of data. Every compressed file and every reliable link rests on this theory.
- Cryptography and secure communication. The notion of perfect secrecy is an information-theoretic statement.
- Machine learning. Cross entropy, relative entropy and mutual information are the standard loss functions and the standard diagnostics of modern neural networks.
- Biology. DNA replication and neural signalling are encoding, transmission and error-correction problems.
- Quantum information and quantum computing, where the classical answers change in very counterintuitive ways that have nevertheless been confirmed experimentally.

> The plan of these first lectures is to state five surprising results and to make each of them plausible. The precise derivations come later in the course, in a more structured and organised presentation.

The five results are:

1. There is a hard limit on lossless compression, it is a single number $H(X)$ attached to the source, and that limit can be approached as closely as we like, **but not more** (Sections 2 and 3).
2. Using a code built for the wrong distribution costs exactly one extra quantity, the **relative entropy**, and nothing else (Section 2).
3. Reliability and speed are not in conflict below a threshold. Up to the channel capacity C we can have fast and reliable transmission over noisy channels **with error probability as small as we wish** (Section 4).
4. A **continuous** signal with bounded bandwidth is determined exactly, not approximately, by a **discrete** set of samples (Section 5).
5. Two separated players sharing a few entangled quantum states (*Bell states for instance*) can win **for sure** a cooperative game that no classical strategy can win more often than 8 times out of 9 (Section 6).

The first four are classical Shannon's results. The fifth belongs to quantum information theory and is included to show that the classical picture can be pushed further within the quantum setting.

All five results are surprising in the sense that they contradict the intuition of most people. And the third one has it in a strong historical sense: Before 1948 most practising engineers believed that driving the error probability to zero forced the transmission rate to zero. That belief was not a lazy guess. It is what every code known at the time did, as we check in Section 4. Shannon proved it false.

2 Surprise, Entropy and the Cost of Being Wrong

2.1 How surprised are we?

Entropy is the central object of information theory. Claude Shannon introduced it in 1948. It measures the uncertainty carried by a random variable, and it turns out to be exactly the answer to the compression question.

We build it from a simpler idea. Suppose we learn that an event A has occurred. How surprised should we be?

Two observations fix the answer almost completely.

1. Surprise should depend on A only through its probability, so $s(A) = f(\mathbb{P}(A))$ for some function f defined on $(0, 1]$, and f should be non-increasing. Rare events are the surprising ones.
2. An event of probability 1 carries no surprise, so $f(1) = 0$, and $f(p)$ should grow without bound as p decreases to 0.

These two conditions still leave several candidates. The third condition is the one that selects the logarithm function.

Probability theory separates itself from measure theory by taking independence into account.

If A and B are independent, then observing both should surprise us by the sum of the two separate surprises, because neither observation changes what we expect of the other.

We therefore require

$$A \text{ and } B \text{ independent} \implies s(A \cap B) = s(A) + s(B). \tag{1}$$

Since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ under independence, condition (1) says that $f(pq) = f(p) + f(q)$ for all admissible p and q. That functional equation has essentially one solution.

Theorem 2.1 (Cauchy's multiplicative functional equation)**.** *Let $f : (0, \infty) \to \mathbb{R}$ be continuous and satisfy*

$$f(xy) = f(x) + f(y) \qquad \text{for all } x, y > 0.$$

Then there is a constant $c \in \mathbb{R}$ with $f(x) = c \log x$ for all $x > 0$.

Imposing f non-increasing forces $c < 0$, and normalising so that an event of probability $1/2$ carries **one unit of surprise** gives $c = -1$ with logarithms in base 2.

Definition 2.2 (Surprise)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A \in \mathcal{F}$ have $\mathbb{P}(A) > 0$. The surprise of observing A is

$$s(A) = -\log_2 \mathbb{P}(A). \tag{2}$$

The unit is the bit when the logarithm has base 2, and the nat when the logarithm is natural.

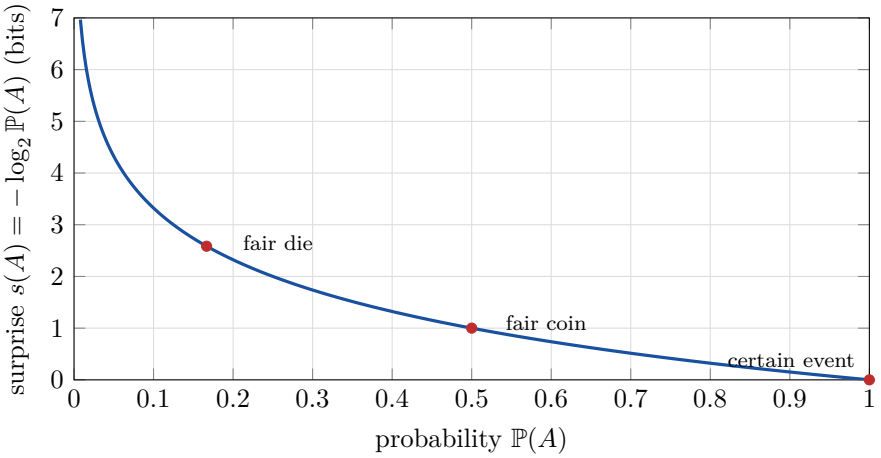


Figure 1: Surprise as a function of probability. A certain event carries no surprise. A fair coin toss carries one bit. A fair die roll carries $\log_2 6 \approx 2.585$ bits.

2.2 Shannon entropy

Entropy is the expected surprise.

Definition 2.3 (Shannon entropy)**.** Let X be a discrete random variable with probability mass function p on a finite or countable alphabet $\mathcal{X}$. The Shannon entropy of X is

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log_2 p(x) = \mathbb{E}[-\log_2 p(X)], \tag{3}$$

with the convention $0 \log_2 0 = 0$.

The convention is the right one because $t \log_2 t \to 0$ as $t \downarrow 0$, so impossible symbols contribute nothing.

Remark. The entropy depends on X only through its distribution, so we also write $H(p)$ when the distribution is what matters.

Theorem 2.4 (Basic properties)**.** *Let X take values in a finite alphabet $\mathcal{X}$.*

1. *$H(X) \geq 0$, with equality if and only if X is constant almost surely.*
2. *$H(X) \leq \log_2 |\mathcal{X}|$, with equality if and only if X is uniform on $\mathcal{X}$.*
3. *If H_b denotes entropy computed with logarithms in base b, then*

$$H_b(X) = (\log_b c)\, H_c(X) = \frac{H_c(X)}{\log_c b}. \tag{4}$$

In particular $H_2(X) = H_e(X)/\ln 2$, so one nat equals $1/\ln 2 \approx 1.4427$ bits.

Example 2.5 (Binary entropy)**.** Let X take the value 0 with probability p and the value 1 with probability $1-p$. Then

$$H(X) = H(p) := -p \log_2 p - (1-p) \log_2 (1-p). \tag{5}$$

The function H is the binary entropy function. It vanishes at $p = 0$ and at $p = 1$, it is symmetric about $p = 1/2$, it is strictly concave, and its maximum value is $H(1/2) = 1$ bit.

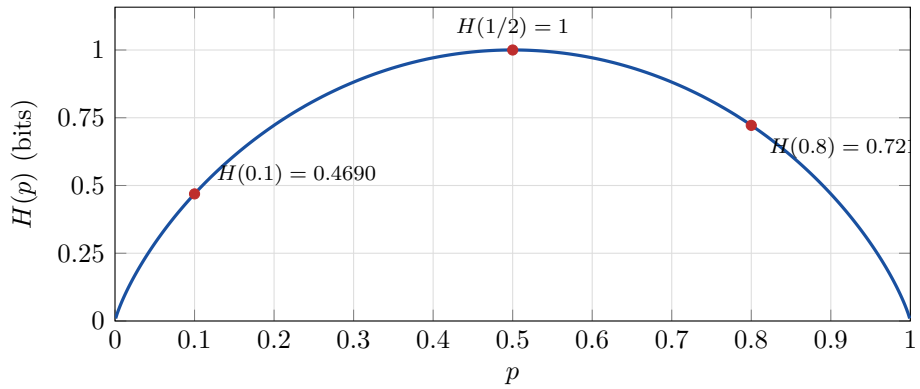


Figure 2: The binary entropy function. The three marked values are used in what follows.

Example 2.6 (Fair coin)**.** For $p = 1/2$,

$$H(X) = -\tfrac{1}{2} \log_2 \tfrac{1}{2} - \tfrac{1}{2} \log_2 \tfrac{1}{2} = \tfrac{1}{2} + \tfrac{1}{2} = 1 \text{ bits}.$$

Example 2.7 (Biased coin)**.** For a coin with $\mathbb{P}(\text{heads}) = 0.8$,

$$H(X) = -0.8 \log_2 0.8 - 0.2 \log_2 0.2 = 0.2575 + 0.4644 = 0.7219 \text{ bits}.$$

The biased coin is more predictable than the fair one, and its entropy is correspondingly smaller.

Remark. (Entropy for continuous random variables)
Everything above is stated for discrete variables. For a random variable Y with density f_Y the *natural candidate* is what is called the **differential entropy** $h(Y) = -\int f_Y \log_2 f_Y$.
It is not the limit of the discrete entropy. Take X_N uniform on $\{1/N, 2/N, \ldots, 1\}$ and Y uniform on $[0,1]$. Then $H(X_N) = \log_2 N \to \infty$, while $h(Y) = 0$. In general $H(X_N) \approx h(Y) + \log_2 N$, so

the discrete entropy diverges and the divergent part $\log_2 N$ depends only on the fineness of the discretisation. Differential entropy is what is left after that infinite constant is discarded. It can be negative, it is not invariant under changes of variable, and it is not the number of bits needed to describe Y, since describing a real number exactly needs infinitely many bits. But we will see that it has several important and useful properties.
Relative entropy and mutual information, by contrast, do pass to the continuous limit unchanged. We return to this later in the course.

2.3 Alice, Bob, and what entropy is counting

Shannon's source coding theorem says that $H(X)$ is exactly the minimum expected number of bits needed to describe one observation of X **losslessly**.
To see what that means, fix a simple concrete situation.

Alice tosses a coin n times and reports the results to Bob, who is nearby and can hear her but cannot see the coin. After each toss she calls out `1` for heads and `0` for tails. We assume the channel is noiseless, meaning Bob always hears correctly whatever Alice calls out. With a fair coin Alice uses n calls to transmit n tosses, which is one bit per toss, and Example 2.7 suggests that a biased coin should need less.

> **First question.** With $p = 0.8$ the entropy is 0.7219 bits per toss. Can Alice really get down to about 0.72 calls per toss on average, for large n, and never lose information?

Now replace the coin by a fair die. Alice observes independent and identically distributed $X_1, \dots, X_n$ uniform on $\{1, 2, 3, 4, 5, 6\}$.

Example 2.8 (Fair die)**.**

$$H(X) = -\sum_{i=1}^{6} \tfrac{1}{6} \log_2 \tfrac{1}{6} = \log_2 6 \approx 2.5850 \text{ bits}. \tag{6}$$

If Alice is allowed to call out one of six distinct sounds, each call carries $\log_2 6$ bits and there is nothing to improve. But suppose she insists on binary calls, which is the situation of every real digital system. Then she and Bob must agree in advance on a code, a map from die outcomes to binary strings.

The simplest choice is a fixed-length code of length 3, since $2^2 = 4$ is too small and $2^3 = 8$ is enough.

die outcome	codeword
1	000
2	001
3	010
4	011
5	100
6	101

Table 1: A fixed-length code for the fair die. The strings 110 and 111 are never used.

Every outcome costs exactly 3 bits, so the expected cost is 3 bits per symbol while the entropy is 2.5850 bits per symbol. Alice is wasting $3 - \log_2 6 \approx 0.4150$ bits on every symbol she sends.

The reason is visible in Table 1. The code is the natural code for a source that is uniform on eight symbols. Alice is using a code built for the uniform distribution on $\{1, \dots, 8\}$ while the true distribution is uniform on $\{1, \dots, 6\}$.

> **Second question.** Using a code designed for the wrong distribution has a cost. Is that cost a well-defined function of the two distributions?

2.4 Cross entropy and relative entropy

Let P be the true distribution and let Q be the distribution the code was built for. A code matched to Q spends about $-\log_2 Q(x)$ bits on the symbol x, which is the surprise that Q assigns to x. Averaging against the true distribution gives the following quantity.

Definition 2.9 (Cross entropy)**.** For distributions P and Q on a discrete set $\mathcal{X}$,

$$H(P, Q) = -\sum_{x \in \mathcal{X}} P(x) \log_2 Q(x). \tag{7}$$

In the die example, take

$$P = \mathrm{Unif}\{1, \dots, 6\}, \qquad Q = \mathrm{Unif}\{1, \dots, 8\},$$

where P is extended to $\{1, \dots, 8\}$ by $P(7) = P(8) = 0$.

Then

$$H(P, Q) = -\sum_{x=1}^{6} \tfrac{1}{6} \log_2 \tfrac{1}{8} = 3 \text{ bits},$$

which is exactly the expected length of the fixed-length code. The excess over the entropy is

$$H(P, Q) - H(P) = 3 - \log_2 6 \approx 0.4150 \text{ bits per symbol.}$$

The next definition names that excess.

Definition 2.10 (Relative entropy)**.** The relative entropy, also called the Kullback-Leibler divergence, of P with respect to Q is

$$D_{\mathrm{KL}}(P \,\|\, Q) = \sum_{x \in \mathcal{X}} P(x) \log_2 \frac{P(x)}{Q(x)}, \tag{8}$$

with the conventions $0 \log_2 \frac{0}{q} = 0$ and $p \log_2 \frac{p}{0} = +\infty$ for $p > 0$.

Theorem 2.11 (Properties of relative entropy)**.**

1. $D_{\mathrm{KL}}(P \,\|\, Q) \geq 0$*, with equality if and only if* $P = Q$*.*
2. *In general* $D_{\mathrm{KL}}(P \,\|\, Q) \neq D_{\mathrm{KL}}(Q \,\|\, P)$*.*
3. D_{KL} *is jointly convex in the pair* (P, Q)*.*

The two quantities are related by an identity that does all the work.

Proposition 2.12 (Decomposition of the coding cost)**.** *For any* P *and* Q *on* $\mathcal{X}$*,*

$$H(P, Q) = H(P) + D_{\mathrm{KL}}(P \,\|\, Q). \tag{9}$$

Proof. Write $\log_2 Q(x) = \log_2 P(x) - \log_2 \frac{P(x)}{Q(x)}$ for every x with $P(x) > 0$, multiply by $-P(x)$ and sum over x. □

Identity (9) answers the second question. The expected cost of coding a source P with a code built for Q splits into two terms. The first term is $H(P)$, which no code can avoid. The second term is $D_{\mathrm{KL}}(P \,\|\, Q)$, which is the entire penalty for using the wrong distribution, and it vanishes exactly when $Q = P$.

Let us verify (9) on the die. With P and Q as above,

$$\begin{aligned} D_{\mathrm{KL}}(P \,\|\, Q) &= \sum_{x=1}^{6} \frac{1}{6} \log_2 \frac{1/6}{1/8} + \sum_{x=7}^{8} 0 \cdot \log_2 \frac{0}{1/8} \\ &= 6 \cdot \frac{1}{6} \cdot \log_2 \frac{8}{6} \\ &= \log_2 \frac{4}{3} \approx 0.4150 \text{ bits}, \end{aligned} \tag{10}$$

which matches $3 - \log_2 6$ exactly, as (9) requires.

> The waste of the fixed-length die code is not an accident of that particular table. It is $D_{\mathrm{KL}}(P \,\|\, Q)$, and any code built for Q pays it.

Additional comment 2.1. The asymmetry is important and not a technicality

The die example makes the asymmetry concrete. We computed $D_{\mathrm{KL}}(P \,\|\, Q) = \log_2 \frac{4}{3}$, a small finite number. Now reverse the roles. Since $Q(7) = 1/8 > 0$ while $P(7) = 0$, the sum defining $D_{\mathrm{KL}}(Q \,\|\, P)$ contains the term $\frac{1}{8} \log_2 \frac{1/8}{0} = +\infty$, so

$$D_{\mathrm{KL}}(Q \,\|\, P) = +\infty.$$

The interpretation is clear. Coding a 6-valued source with an 8-symbol code wastes a fraction of a bit. Coding an 8-valued source with a code that has no codeword at all for two of the symbols is impossible, and the infinite value records this.

Additional comment 2.2. Relative entropy is not a distance, and what replaces that

Two of the three metric axioms fail. It is not symmetric, as the previous comment shows in the strongest possible way, and it does not satisfy the triangle inequality. What survives is enough for most purposes. Property 1 of Theorem 2.11 makes D_{KL} a divergence, meaning it separates points, and Pinsker's inequality

$$D_{\mathrm{KL}}(P \,\|\, Q) \;\geq\; \frac{2}{\ln 2} \, \|P - Q\|_{\mathrm{TV}}^2$$

bounds a genuine metric, the total variation distance, from above by the divergence. So small relative entropy does force the distributions to be close, even though the converse fails badly.

Additional comment 2.3. A notational warning

Some texts write $H(P;Q)$ or $H^{\times}(P,Q)$ for the cross entropy. The notation $H(P,Q)$ used here collides with the standard notation $H(X,Y)$ for the joint entropy of two random variables, which is a different object. The arguments disambiguate in practice, since cross entropy takes two distributions on the same alphabet and joint entropy takes two random variables, but it is worth keeping in mind.

Additional comment 2.4. Why machine learning is full of cross entropy

A classifier outputs a distribution Q_θ over labels and is trained on data drawn from an unknown true distribution P. Minimising the empirical cross entropy $H(P, Q_\theta)$ over θ is, by (9), the same as minimising $D_{\mathrm{KL}}(P \,\|\, Q_\theta)$, because $H(P)$ does not depend on θ. So the standard training loss of a neural network is literally the number of extra bits the model wastes relative to a perfect model.

3 A Better Code, and the Limit of All Codes

We left two questions open. The fixed-length die code wastes 0.4150 bits per symbol, and we want to know how much of that waste is avoidable. We also want to know whether the biased coin can really be sent at about 0.72 calls per toss.

3.1 Intuition: the twenty questions game

In the game of twenty questions we identify an unknown object using yes-or-no questions. A good player splits the remaining possibilities as evenly as possible at each step, because a question whose two answers are equally likely removes a full bit of uncertainty, while a lopsided question usually removes much less.

A strategy for that game is a binary decision tree. Reading off the answers along the path to an object gives a binary string, so a strategy is the same thing as a binary code. Playing well means building a tree whose average depth, weighted by the probabilities of the objects, is as small as possible. That is exactly the coding problem, and Huffman's algorithm solves it.

3.2 Huffman's construction for the fair die

Huffman's algorithm builds the tree from the leaves upward. At each step it merges the two least probable nodes into a single node whose probability is the sum. Merging the least probable pair last-but-deepest is what pushes rare symbols to long codewords and frequent symbols to short ones.

Huffman construction for the fair die

1. Start with six nodes, one per outcome, each of probability $\frac{1}{6}$.
2. Merge 1 and 2 into a node of probability $\frac{1}{3}$.
3. Merge 3 and 4 into a node of probability $\frac{1}{3}$.
4. Merge 5 and 6 into a node of probability $\frac{1}{3}$.
5. Merge the nodes $\{1, 2\}$ and $\{3, 4\}$ into a node of probability $\frac{2}{3}$.
6. Merge $\{1, 2, 3, 4\}$ and $\{5, 6\}$ into the root, of probability 1.

Labelling the two branches out of each node by 0 and 1 gives the code.

The code in Table 2 has a property that Table 1 also had but that variable-length codes can easily lose.

Definition 3.1 (Prefix code)**.** A set of binary codewords is a prefix code when no codeword is a prefix of another codeword.

Prefix codes are exactly the codes whose codewords are the leaves of a binary tree. They are uniquely decodable and they can be decoded on the fly, since the decoder recognises the end of

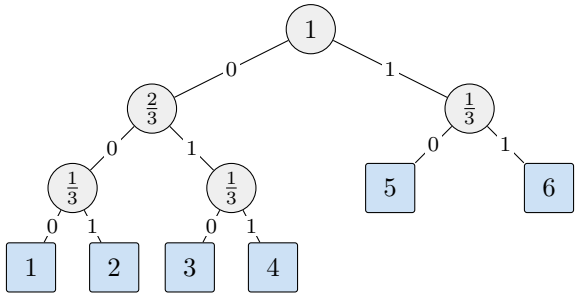


Figure 3: The Huffman tree for the fair die. Outcomes 5 and 6 sit at depth 2 and the other four sit at depth 3.

die outcome	probability	codeword
1	$\frac{1}{6}$	000
2	$\frac{1}{6}$	001
3	$\frac{1}{6}$	010
4	$\frac{1}{6}$	011
5	$\frac{1}{6}$	10
6	$\frac{1}{6}$	11

Table 2: The Huffman code for the fair die.

a codeword the moment it arrives and never needs to look ahead. Reading 01111000 from left to right gives 011, then 11, then 000, that is the rolls 4, then 6, then 1, with no ambiguity at any point.

3.3 How good is the Huffman code?

The expected length of the code in Table 2 is

$$\mathbb{E}\left[L_{\text{Huffman}}\right] = 4 \cdot \frac{1}{6} \cdot 3 + 2 \cdot \frac{1}{6} \cdot 2 = \frac{12}{6} + \frac{4}{6} = \frac{16}{6} = \frac{8}{3} \approx 2.6667 \text{ bits.} \tag{11}$$

The entropy of the source is $\log_2 6 \approx 2.5850$ bits, so the excess is

$$\mathbb{E}\left[L_{\text{Huffman}}\right] - H(P) = \frac{8}{3} - \log_2 6 \approx 0.0817 \text{ bits per symbol.} \tag{12}$$

Compare with the fixed-length code, which wasted 0.4150 bits per symbol. The Huffman code removes about 80 percent of the waste. It does not remove all of it.

> No code that assigns a whole number of bits to each individual outcome can do better than (11) for the fair die. Huffman's algorithm is optimal in that class. Yet 2.6667 is still above 2.5850. The gap is caused by the requirement that each codeword length be an integer.

3.4 Coding blocks, and Shannon's source coding theorem

The way past the integer constraint is to stop coding one symbol at a time. Alice waits for n die rolls and encodes the whole block $(X_1, \ldots, X_n)$ as a single symbol from an alphabet of size 6^n. The integer rounding is then paid once per block instead of once per symbol, so its cost per symbol is divided by n.

Theorem 3.2 (Shannon's source coding theorem)**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed with distribution p on a finite alphabet. Let L_n^* be the minimum expected length of a prefix code for the block $(X_1, \ldots, X_n)$. Then*

$$H(X) \;\le\; \frac{L_n^*}{n} \;<\; H(X) + \frac{1}{n} \qquad \text{for every } n \ge 1, \tag{13}$$

so $L_n^*/n \to H(X)$. Moreover no uniquely decodable code has expected length per symbol below $H(X)$.

> The information in X cannot be squeezed into fewer than $H(X)$ bits per symbol, and it can be squeezed into as close to $H(X)$ bits per symbol as we please.

Both halves of (13) matter. The lower bound is a hard impossibility statement. The upper bound is a construction, and applying Huffman's algorithm to blocks already realises it.

Table 3 shows what happens for the fair die.

block length n	alphabet size 6^n	L_n^*/n (bits)	excess over $\log_2 6$
1	6	2.66667	0.08170
2	36	2.61111	0.02615
3	216	2.60494	0.01998
4	1296	2.60494	0.01998
5	7776	2.58930	0.00434

Table 3: Huffman coding applied to blocks of n fair die rolls. The entropy is $\log_2 6 \approx 2.58496$ bits per roll.

The same computation answers the first question, about the biased coin.

block length n	L_n^*/n (bits)	excess over $H(0.8)$
1	1.00000	0.27807
2	0.78000	0.05807
3	0.72800	0.00607
4	0.74080	0.01887
5	0.73792	0.01599
6	0.72525	0.00332
12	0.72504	0.00311

Table 4: Huffman coding applied to blocks of n tosses of a coin with $\mathbb{P}(\text{heads}) = 0.8$. The entropy is $H(0.8) \approx 0.72193$ bits per toss.

At $n = 1$ Alice needs one call per toss and gains nothing. At $n = 3$ she is already down to 0.728 calls per toss. The bound $H(X) + 1/n$ of (13) is respected at every line, and the limit is $H(0.8) = 0.7219$.

> **Answer to the first question.** Yes. Alice can send the biased coin at an average of about 0.72 calls per toss, and she cannot do better. The apparent unit of one call per toss was never a real limit. It was an artefact of insisting on one codeword per toss.

Additional comment 3.1. Why L_n^*/n is not monotone in n

Table 4 looks wrong at first sight. Going from $n = 3$ to $n = 4$ makes the code worse, from 0.72800 to 0.74080. This is real and not a numerical error. Theorem 3.2 guarantees only the envelope $H \le L_n^*/n < H + 1/n$, and inside that envelope the sequence wanders. The reason is arithmetic. Huffman's algorithm is optimal for each fixed block length, but how well the optimal integer lengths match the ideal lengths $-\log_2 p$ depends on the fine structure of the probabilities $p^k(1-p)^{n-k}$, and that structure changes irregularly with n. The same effect appears at $n = 4$ for the die, where the value repeats instead of improving. What is guaranteed is the limit, and the envelope forces it.

Additional comment 3.2. Kraft's inequality, the reason integers are the obstruction

A prefix code with codeword lengths $\ell_1, \dots, \ell_m$ exists if and only if

$$\sum_{i=1}^{m} 2^{-\ell_i} \;\le\; 1.$$

If we drop the requirement that the ℓ_i be integers and minimise $\sum_i p_i \ell_i$ subject to Kraft's inequality, a Lagrange multiplier computation gives the optimum $\ell_i = -\log_2 p_i$ with value exactly $H(P)$. So the entropy is the answer to the relaxed problem, and the excess in (12) is purely the price of rounding to integers. For the fair die the ideal length is $\log_2 6 = 2.585$ for every outcome, and no integer sits there. Blocking makes the ideal lengths large, and rounding a large number costs proportionally less.

3.5 What if the distribution is unknown?

Huffman's algorithm needs the probabilities. Real files do not come with a probability model attached. Suppose you have a large file and a memory stick that is slightly too small.

Remarkably, the entropy limit can be reached without knowing the source. A universal compressor reads the data, discovers its statistics as it goes, and achieves an average length per symbol converging to the entropy of any stationary ergodic source that produced it, without being told anything in advance. The Lempel-Ziv family of algorithms does this, and it is what runs inside `gzip` and `zip`. This is another surprise, and we return to it later in the course.

3.6 From compression to error correction

Everything so far concerned a noiseless link. Alice and Bob were only trying to be brief. Suppose instead that the connection is bad and that some of what Alice says arrives wrong. Now brevity is dangerous.

> Compression removes redundancy in order to save space.
> Error correction adds redundancy in order to survive noise.

The two goals pull in opposite directions, and the tension between them is the subject of the next section. Compression identifies patterns and repetition and strips them out, so that the original data can be reconstructed exactly, as in Huffman coding, or approximately, as in JPEG and MP3. Error correction inserts carefully structured redundancy so that corrupted symbols can be detected and repaired at the receiver without asking for a retransmission.

Stated that way, the trade-off looks unavoidable. The next section shows that it is not.

4 Reliable Communication over a Noisy Channel

Now the link through which the information is transmitted is unreliable. Think of a telephone line with static, a wireless connection with interference, or a magnetic disk whose bits flip. We want two things at once.

> *Reliability.* The probability that the receiver reconstructs the wrong message should be small.
> *Rate.* The number of message bits carried per use of the channel should be large.

Before stating the theorem we build the *natural* intuition, the intuition that everybody builds, and then see that it is (amazingly) wrong.

4.1 Channels and the binary symmetric channel

Let the input X take values in a finite alphabet $\mathcal{X}$ and let the output Y take values in a finite alphabet $\mathcal{Y}$. A discrete memoryless channel is specified by a conditional distribution

$$p(y \mid x), \qquad x \in \mathcal{X},\ y \in \mathcal{Y},$$

applied independently to each transmitted symbol.

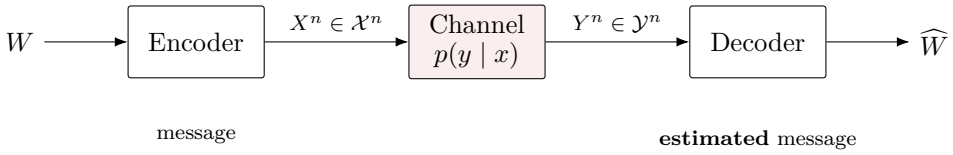


Figure 4: A communication system. The encoder turns a message W into a codeword X^n, the channel produces Y^n, and the decoder produces an estimate $\widehat{W}$.

Example 4.1 (Binary symmetric channel)**.** The binary symmetric channel with crossover probability p, written BSC(p), has $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and

$$p(y \mid x) = \begin{cases} 1 - p, & y = x, \\ p, & y \neq x. \end{cases}$$

Each transmitted bit is flipped independently with probability p.

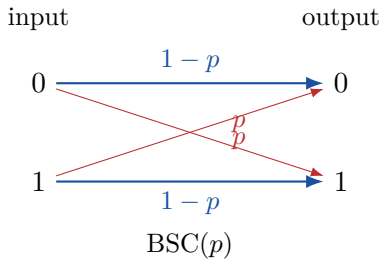


Figure 5: The binary symmetric channel with crossover probability p.

Throughout this section we fix $p = 0.1$. One bit in ten is flipped, which is a very noisy channel by any practical standard.

4.2 The natural idea: repetition

If noise might corrupt a bit, send it several times. The three-fold repetition code R_3 is

$$0 \mapsto 000, \qquad 1 \mapsto 111,$$

and the receiver decodes by majority vote. The vote is wrong exactly when two or three of the three bits are flipped, so

$$P_{\mathrm{e}}(R_3) = \binom{3}{2} p^2(1-p) + \binom{3}{3} p^3 = 3(0.01)(0.9) + 0.001 = 0.028. \tag{14}$$

The error probability drops from 0.1 to 0.028. The price is that we need three channel uses now to carry one message bit, so the rate is $R = \frac{1}{3}$.

For more reliability, repeat more.

The five-fold code R_5 corrects up to two errors and has

$$P_{\mathrm{e}}(R_5) = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k} = 0.00856,$$

at rate $R = \frac{1}{5}$.

The pattern is clear and it is discouraging.

> More reliability requires more redundancy.
> More redundancy means a lower rate.
> Driving the error probability to zero appears to drive the rate to zero as well.

This is quantitative, not vague. Suppose we insist on $P_{\mathrm{e}} < 10^{-15}$, which is a routine requirement for stored data. The shortest repetition code that reaches it over BSC(0.1) is R_{63}, with $P_{\mathrm{e}} \approx 3.9 \times 10^{-16}$ and rate $\frac{1}{63}$. To store 10 gigabytes of data this way we would need 630 gigabytes of disk.

4.3 Structured redundancy does better, but still loses

Repetition is wasteful because every repeated bit carries the same information. A better code lets each redundant bit check a different combination of message bits.

Example 4.2 (Hamming $(7,4)$ code)**.** Four message bits (d_1, d_2, d_3, d_4) are extended by three parity bits to a codeword of length 7,

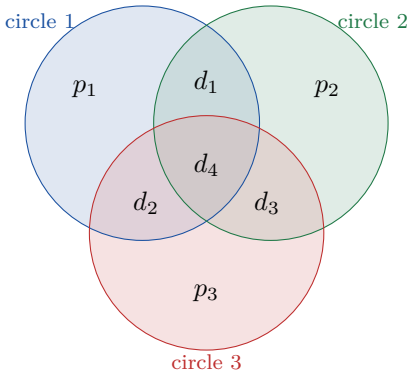
$$(d_1, d_2, d_3, d_4) \;\longmapsto\; (d_1, d_2, d_3, d_4, p_1, p_2, p_3),$$

so the rate is $R = \frac{4}{7}$. Any single bit error produces a distinct pattern of failed parity checks and is therefore located and repaired.

The block is decoded wrongly when two or more of the seven bits are flipped, so

$$\begin{aligned} P_{\mathrm{block}} &= 1 - \left[\binom{7}{0} (0.1)^0 (0.9)^7 + \binom{7}{1} (0.1)^1 (0.9)^6 \right] \\ &= 1 - [0.4783 + 0.3720] \\ &= 0.1497. \end{aligned} \tag{15}$$

It seems that for better reliability , we must pay with rate. The trade-off looks like a law of nature.



$$p_1 = d_1 \oplus d_2 \oplus d_4,$$
$$p_2 = d_1 \oplus d_3 \oplus d_4,$$
$$p_3 = d_2 \oplus d_3 \oplus d_4.$$

Figure 6: The Hamming $(7, 4)$ code. Each circle contains four bits and its parity bit is chosen so that the four bits inside it sum to zero modulo two. A single flipped bit is identified by the set of circles whose parity fails.

> **The intuition everybody had before 1948:** There is an unavoidable minimum error probability at any positive rate, and error probability can be made arbitrarily small only by letting the rate go to zero.

This intuition is wrong.

4.4 Shannon's channel coding theorem

In 1948 Shannon proved that the behaviour is not a smooth trade-off at all. We have a phase transition, of the kind familiar from statistical mechanics, in the parameter R. There is a critical value C, called the capacity of the channel, and the two sides of C behave completely differently.

Theorem 4.3 (Shannon's channel coding theorem)**.** *Every discrete memoryless channel has a capacity $C \geq 0$ with the following two properties.*

1. Achievability. *For every rate $R < C$ and every $\varepsilon > 0$ there are codes of rate at least R and block length large enough whose probability of decoding error is below ε.*
2. Converse. *For every rate $R > C$ the probability of decoding error is bounded away from zero, no matter how the code and the decoder are chosen and no matter how long the blocks are.*

> Below the capacity, reliability and rate are not in conflict. We may fix any rate $R < C$, keep it fixed, and still manage to have an error probability as close to zero as we want.

For the binary symmetric channel the capacity is

$$C = 1 - H(p), \tag{16}$$

where H is the binary entropy function of Figure 2. Every channel model has such a formula. A few are collected in Table 5.

4.5 What the theorem says about our examples

At $p = 0.1$,

$$H(0.1) = 0.4690, \qquad C = 1 - H(0.1) = 0.5310 \text{ bits per channel use.}$$

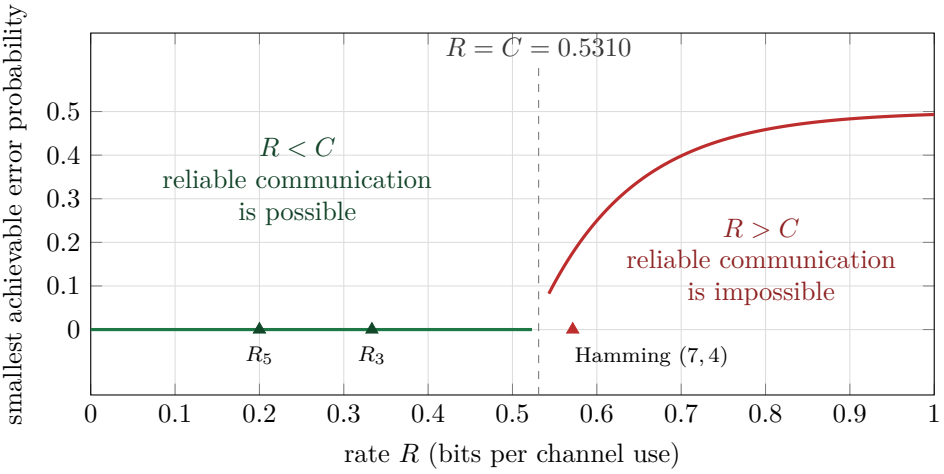


Figure 7: The phase transition at the capacity, drawn for BSC(0.1) where $C = 0.5310$. To the left of C the smallest achievable error probability is zero. To the right it is bounded away from zero.

channel	input	output	capacity C
binary symmetric, BSC(p)	binary	binary	$1 - H(p)$
binary erasure, BEC(ε)	binary	ternary	$1 - \varepsilon$
Z-channel	binary	binary	$\log_2\left(1 + (1-p)\, p^{p/(1-p)}\right)$
additive white Gaussian noise	$\mathbb{R}$	$\mathbb{R}$	$\frac{1}{2}\log_2(1 + \mathrm{SNR})$

Table 5: Capacities of some standard channels.

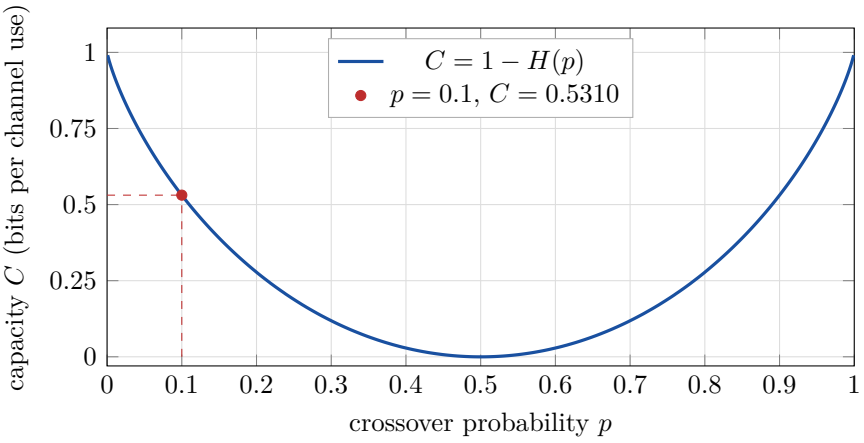


Figure 8: Capacity of the binary symmetric channel as a function of the crossover probability.

- The Hamming $(7, 4)$ code has rate $4/7 = 0.5714$, which is *above* $C = 0.5310$. The converse in Theorem 4.3 says that at this rate no code and no decoder can be reliable, so the failure of Hamming $(7, 4)$ here is not a defect of that particular code.
- The repetition codes have rates $1/3$ and $1/5$, both well below C. Achievability says that codes of those rates exist with error probability as small as we like. Repetition does not achieve this, but something else does.
- Any rate below 0.5310 is usable with arbitrarily high reliability. The gap between 0.5310 and what the codes in the table deliver is entirely a gap in our construction, not in the channel.

The disk example. Store 10 gigabytes on a medium that behaves like BSC(0.1), with a required error probability below 10^{-15}. Repetition needs 630 gigabytes. Shannon's theorem says that $10/C = 18.83$ gigabytes suffice. A disk of twice the nominal size is enough.

4.6 What Shannon does not tell us

The proof gives no useful description of a good code!

This is more surprising than the theorem. Shannon's argument is a random coding argument. Instead of constructing a code, he considers the ensemble of all codes of a given rate with codewords drawn at random, computes the average error probability over the ensemble, and shows that this average tends to zero when $R < C$. Since the average is small, at least one code in the ensemble is at least as good as the average, so good codes exist.

An abstract existence proof without any clue on how to construct such a code!!

The argument is not constructive. It proves existence without exhibiting a single example, and the codes it produces have no structure that would make encoding and decoding practical. Closing that gap took about fifty years and produced codes that now run inside wireless standards, satellite links and storage systems, and they operate within a fraction of a decibel of the Shannon limit.

Additional comment 4.1. Where the number C comes from

The capacity is not postulated. It is

$$C = \max_{p_X} I(X;Y),$$

the maximum over input distributions of the mutual information between input and output. Mutual information is built from the quantities of Section 2 through $I(X;Y) = H(Y) - H(Y \mid X)$. For the binary symmetric channel the noise is independent of the input, so $H(Y \mid X) = H(p)$ for every input distribution, and $H(Y) \leq 1$ with equality when the input is uniform. This gives $C = 1 - H(p)$ immediately. The general definition and the proof that this particular maximum is the operational threshold are the content of the coding theorem.

Additional comment 4.2. Why the converse is the harder half to believe

Achievability is surprising but at least it is an existence statement, and random coding makes it plausible. The converse is a statement about every possible code, including codes nobody has thought of. It follows from a chain of inequalities built on the data processing inequality, which says that no processing of Y can increase the information it carries about X. That single inequality, applied along the chain $W \to X^n \to Y^n \to \widehat{W}$, is what forbids clever decoders from beating the capacity.

5 Continuous Signals and the Sampling Theorem

The next surprise concerns continuous signals such as music, speech and analog measurements. How much of such a signal survives when we record only finitely many numbers per second?

5.1 The objection

The natural way to send a continuous signal through a digital system is sampling. We record the value of the signal at regular instants and transmit the resulting sequence of numbers.

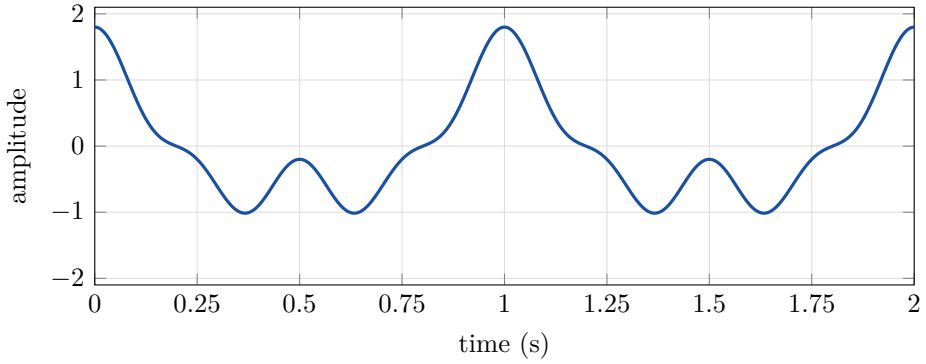


Figure 9: A signal $f(t) = \cos(2\pi t) + \frac{1}{2}\cos(4\pi t) + \frac{3}{10}\cos(8\pi t)$, built from cosines at 1 Hz, 2 Hz and 4 Hz.

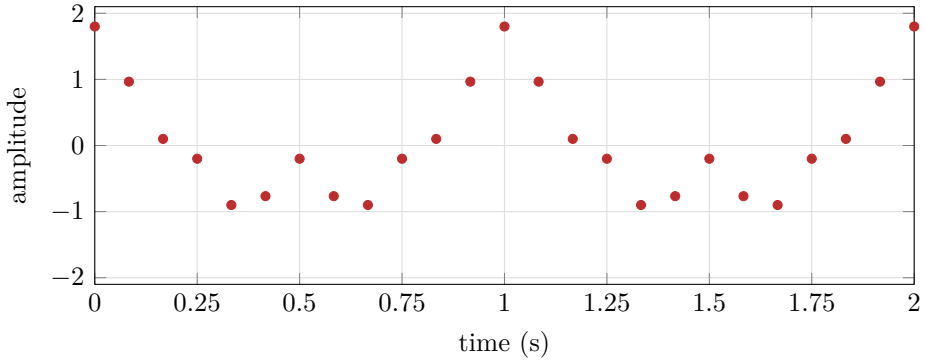


Figure 10: All that is transmitted. The samples of f taken at 12 Hz.

Comparing the two figures raises an obvious objection. Between two consecutive samples the signal could have done anything. Uncountably many functions pass through any finite set of points, so we appear to have thrown away an infinite amount of information. Reconstruction should be impossible, and digital audio should be intrinsically inferior to an analog recording.

5.2 The missing constraint

The objection is correct about arbitrary functions and wrong about signals. Real signals are not arbitrary. They are constrained by the physics that produced them and by the physiology that receives them.

- A physical source cannot produce components of arbitrarily high frequency. Producing them would require unbounded energy or unbounded acceleration.

- Human hearing responds to frequencies from about 20 Hz to about 20 kHz. Content above that range is inaudible.

This suggests restricting attention to signals with bounded bandwidth.

Definition 5.1 (Band-limited signal)**.** A signal f is band-limited to B hertz when its Fourier transform F satisfies $F(\nu) = 0$ for all $|\nu| > B$.

The signal of Figure 9 is band-limited to $B = 4$ Hz, since it contains only the frequencies 1, 2 and 4.

5.3 The Nyquist-Shannon sampling theorem

Theorem 5.2 (Nyquist-Shannon sampling theorem)**.** *Let f be band-limited to B hertz and let the sampling rate satisfy $f_s > 2B$, equivalently $T = 1/f_s < 1/(2B)$. Then f is determined by the samples $\{f(nT)\}_{n\in\mathbb{Z}}$, and it is recovered by*

$$f(t) = \sum_{n\in\mathbb{Z}} f(nT)\,\operatorname{sinc}\!\left(\frac{t - nT}{T}\right), \qquad \operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}. \tag{17}$$

> There is exactly one band-limited signal consistent with the samples. The samples do not approximate f. They determine f.

The rate $2B$ is called the Nyquist rate. For the signal of Figure 9 it is 8 Hz, and the samples in Figure 10 were taken at 12 Hz, comfortably above it.

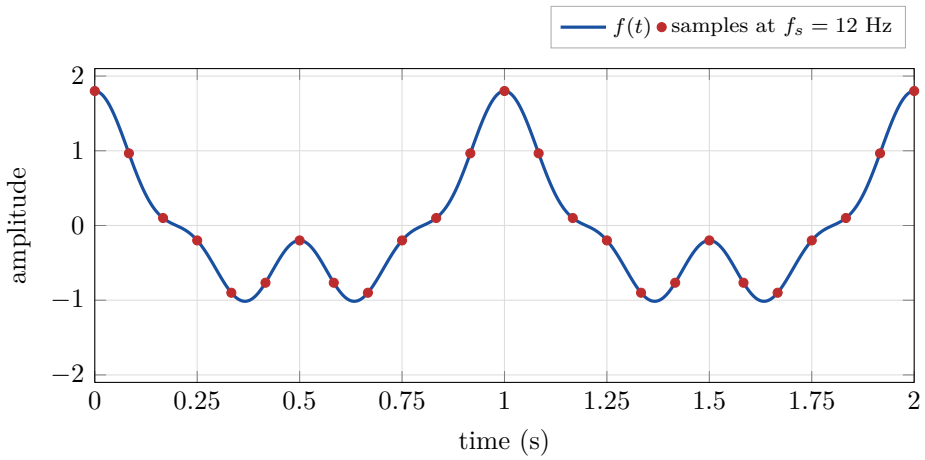


Figure 11: The signal and its samples together. Every value of the blue curve is computable from the red dots through (17).

This is the surprise. An object with uncountably many degrees of freedom, a function on an interval, is pinned down exactly by a countable list of numbers arriving at a finite rate. The band limit converts an infinite-dimensional problem into a finite-rate one.

5.4 What happens below the Nyquist rate

The condition $f_s > 2B$ is not decoration. Below it, distinct band-limited signals produce identical samples and reconstruction is impossible in principle.

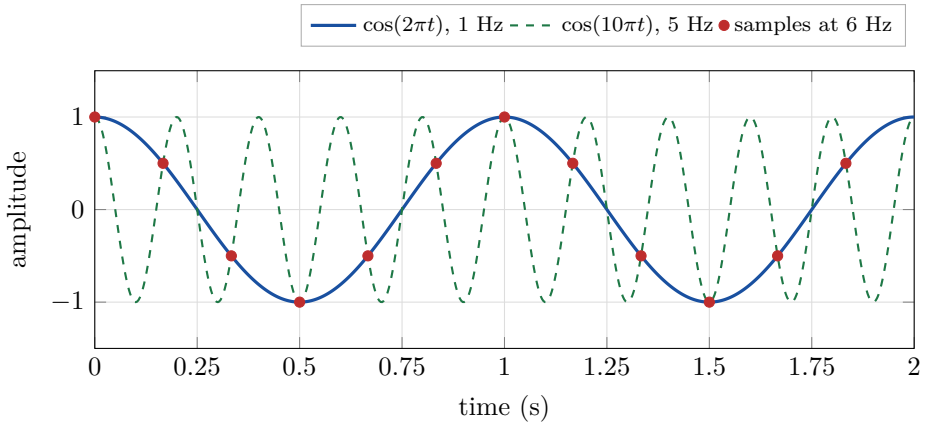


Figure 12: Sampling at 6 Hz. The 1 Hz cosine and the 5 Hz cosine agree at every sample point, so the samples cannot distinguish them. At 6 Hz the theorem covers only signals band-limited to below 3 Hz, and the 5 Hz signal violates that.

The two curves in Figure 12 share every sample because $\cos(10\pi n/6) = \cos(2\pi n/6)$ for every integer n. The high frequency masquerades as a low one. This effect is called aliasing, and it is why every real digitiser applies an analog low-pass filter before sampling.

5.5 Consequences

Theorem 5.2 is the theoretical basis of digital signal processing. The compact disc samples at 44.1 kHz, which exceeds twice the 20 kHz upper limit of human hearing, so everything a listener can hear is inside the band that the theorem covers.

Additional comment 5.1. Counting degrees of freedom

Here is why $2B$ is the right number and not an arbitrary threshold. A real signal band-limited to B and observed over a long window of length T is determined by its Fourier coefficients at the frequencies in $[-B, B]$, spaced $1/T$ apart. That is about $2BT$ real numbers, since the coefficients at ν and $-\nu$ are conjugate for a real signal. Sampling at rate f_s over the same window produces f_sT numbers. Matching the two counts gives $f_s = 2B$. Below that rate we are collecting fewer numbers than the signal has degrees of freedom, so information must be lost, and Figure 12 shows it being lost.

Additional comment 5.2. Strictly band-limited signals do not exist

Theorem 5.2 is an idealisation, and it is worth knowing exactly how. A signal whose Fourier transform vanishes outside a bounded set extends to an entire function of exponential type. Such a function cannot vanish on any interval without vanishing identically. So a strictly band-limited signal has infinite duration, and no signal that starts and stops is band-limited. In practice one sacrifices a little of each. The anti-alias filter attenuates rather than annihilates the high frequencies, the sampling rate is chosen with a margin above twice the nominal band edge, and the reconstruction error is made small rather than zero. The 44.1 kHz rate carries a 10 percent margin above 40 kHz for exactly this reason.

Additional comment 5.3. Sampling is exact, storage is not

The theorem says the samples determine the signal. It says nothing about writing the samples down. Each sample is a real number and needs infinitely many bits to record exactly, so every real system quantises, for instance to 16 bits per sample on a compact disc. Quantisation, not sampling, is the

source of the residual error in digital audio. The correct statement is therefore that the sampling step loses nothing and the quantisation step loses a controlled amount. Confusing the two is the usual source of the claim that digital audio is intrinsically inferior.

Additional comment 5.4. The bridge back to channel capacity

The sampling theorem and Section 4 meet in the Shannon-Hartley formula. A channel of bandwidth B with signal-to-noise ratio SNR has capacity

$$C = B \log_2(1 + \mathrm{SNR}) \text{ bits per second.}$$

The derivation samples the waveform at the Nyquist rate $2B$, which turns the continuous channel into $2B$ uses per second of the additive Gaussian channel of Table 5, each of capacity $\frac{1}{2}\log_2(1+\mathrm{SNR})$. Multiplying gives the formula. This is why bandwidth and noise, and nothing else, determine how fast a physical link can carry information.

6 The Mermin-Peres Magic Square

The last surprise is not Shannon's. It belongs to quantum information theory, and it shows that the classical picture of separated parties sharing a strategy is incomplete.

Quantum mechanics does not agree with intuition, and there is still genuine disagreement about how to interpret it. What is not in dispute is that its predictions have been confirmed experimentally to remarkable precision. The example below is one of many that could have been chosen. It is chosen because the classical side is a two-line parity argument that anyone can check, and the quantum side is a short computation with 4×4 matrices.

6.1 The game

Two players, Alice and Bob, face a referee.

1. Before the game starts, Alice and Bob may agree on anything and may share any physical resources they like.
2. Once the game starts they are separated and cannot communicate. The separation is enforced by distance, so that relativity forbids any signal from reaching the other player before the answers are given.
3. The referee picks a row index $i \in \{1, 2, 3\}$ uniformly at random and sends it to Alice, and independently picks a column index $j \in \{1, 2, 3\}$ uniformly at random and sends it to Bob.
4. Alice returns three values in $\{+1, -1\}$, one for each cell of row i. Bob returns three values in $\{+1, -1\}$, one for each cell of column j.
5. They win when all three of the following hold.
   (a) The product of Alice's three values is $+1$.
   (b) The product of Bob's three values is -1.
   (c) Alice and Bob assign the same value to the cell (i, j) where the row and the column meet.

Classical players cannot win more often than 8 times out of 9. Players sharing an entangled quantum state win every time.

6.2 No classical strategy wins always

Suppose Alice and Bob agree in advance on a full table of values $v_{ij} \in \{+1, -1\}$ for the nine cells, and each answers by reading off the requested row or column. Condition (c) is then automatic. Conditions (a) and (b) require

$$\prod_{j=1}^{3} v_{ij} = +1 \quad \text{for } i = 1, 2, 3, \qquad \prod_{i=1}^{3} v_{ij} = -1 \quad \text{for } j = 1, 2, 3. \tag{18}$$

Multiply all nine entries together, grouping first by rows and then by columns:

$$\prod_{i=1}^{3}\prod_{j=1}^{3} v_{ij} = \prod_{i=1}^{3}\left(\prod_{j=1}^{3} v_{ij}\right) = (+1)^3 = +1, \tag{19}$$

$$\prod_{i=1}^{3}\prod_{j=1}^{3} v_{ij} = \prod_{j=1}^{3}\left(\prod_{i=1}^{3} v_{ij}\right) = (-1)^3 = -1. \tag{20}$$

The same product equals $+1$ and -1, which is impossible. Hence no table satisfies (18), and no classical strategy wins with certainty.

The obstruction is a parity count and does not depend on the values chosen. It is the whole classical content of the game.

6.3 The best classical strategy

Since they cannot satisfy all nine constraints, Alice and Bob satisfy eight of them and disagree on one cell. Take the table

$$\begin{pmatrix} +1 & +1 & +1 \\ +1 & +1 & +1 \\ -1 & -1 & * \end{pmatrix} \tag{21}$$

with the cell $(3, 3)$ left open. Rows 1 and 2 already have product $+1$. Columns 1 and 2 already have product -1. If Alice is asked for row 3 she answers $(-1, -1, +1)$, whose product is $+1$. If Bob is asked for column 3 he answers $(+1, +1, -1)$, whose product is -1. Each of them is always consistent with the rule they must satisfy.

They disagree only at cell $(3, 3)$, and that cell is queried only when the referee picks $i = 3$ and $j = 3$, which happens with probability $\frac{1}{9}$. Therefore this strategy wins with probability $\frac{8}{9}$, and this is optimal.

Additional comment 6.1. Randomised strategies do not help

The argument above bounds deterministic strategies. A general classical strategy allows Alice and Bob to share a random variable Λ agreed in advance and to answer as deterministic functions of Λ and their own question. Conditionally on $\Lambda = \lambda$ the strategy is deterministic, so its winning probability is at most $\frac{8}{9}$. The overall winning probability is the average of these conditional probabilities and is therefore also at most $\frac{8}{9}$. Shared randomness never beats the best deterministic strategy in a game of this type, because the winning probability is affine in the shared distribution. This is why the quantum advantage cannot be explained by any classical correlation, however elaborate.

$+1$	$+1$	$+1$	row product $= +1$
$+1$	$+1$	$+1$	row product $= +1$
-1	-1	$+1$ -1	row product $= +1$ (Alice's row 3)
-1	-1	-1	Bob's column 3
column products			

Figure 13: The two players agree everywhere except at the cell $(3, 3)$. Alice must write $+1$ there and Bob must write -1 there, so they lose exactly when the referee asks for row 3 and column 3.

6.4 The quantum strategy

Alice and Bob now use a quantum resource. The state they share is built from Bell states. The Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \tag{22}$$

is the simplest entangled state of two qubits. It is the state behind what Einstein called spooky action at a distance.

Resources. Alice and Bob prepare two copies of $|\Phi^+\rangle$. Alice keeps the first qubit of each copy, calling them A_1 and A_2. Bob keeps the second qubit of each copy, calling them B_1 and B_2. They also carry identical copies of a table which tells them, for each cell of the square, which measurement to perform. Then they separate.

The table. Each cell of the square holds a two-qubit observable, built from the Pauli matrices

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{23}$$

arranged as

$$\mathcal{O} = \begin{pmatrix} \sigma_X \otimes \mathbb{I} & \mathbb{I} \otimes \sigma_X & \sigma_X \otimes \sigma_X \\ \mathbb{I} \otimes \sigma_Z & \sigma_Z \otimes \mathbb{I} & \sigma_Z \otimes \sigma_Z \\ -\sigma_X \otimes \sigma_Z & -\sigma_Z \otimes \sigma_X & \sigma_Y \otimes \sigma_Y \end{pmatrix}, \tag{24}$$

where $\mathbb{I}$ is the 2×2 identity. Every entry O_{ij} is Hermitian and satisfies $O_{ij}^2 = \mathbb{I} \otimes \mathbb{I}$, so its eigenvalues are $+1$ and -1. These are the two possible measurement outcomes, which is exactly the set of values the game asks for.

Play. Alice receives i, measures the three observables O_{i1}, O_{i2}, O_{i3} on her pair (A_1, A_2), and reports the three outcomes. Bob receives j, measures O_{1j}, O_{2j}, O_{3j} on his pair (B_1, B_2), and reports the three outcomes.

6.5 Why this wins with certainty

Three algebraic facts about (24) do all the work.

Proposition 6.1 (Properties of the square)**.** *For the array $\mathcal{O}$ of (24):*

1. *any two entries in the same row commute, and any two entries in the same column commute,*
2. *the product of the three entries of each row is $+\mathbb{I}\otimes\mathbb{I}$,*
3. *the product of the three entries of each column is $-\mathbb{I}\otimes\mathbb{I}$.*

Sketch. Two tensor products of Pauli matrices commute when they anticommute in an even number of the two factors. Every pair within a row or within a column of $\mathcal{O}$ has this property. For the products, the first two rows use $\sigma_X^2=\sigma_Z^2=\mathbb{I}$ directly. For row 3,

$$(-\sigma_X\otimes\sigma_Z)(-\sigma_Z\otimes\sigma_X)(\sigma_Y\otimes\sigma_Y)=(\sigma_X\sigma_Z\sigma_Y)\otimes(\sigma_Z\sigma_X\sigma_Y)=(-i\,\mathbb{I})\otimes(+i\,\mathbb{I})=\mathbb{I}\otimes\mathbb{I},$$

using $\sigma_X\sigma_Z=-i\sigma_Y$ and $\sigma_Z\sigma_X=+i\sigma_Y$ together with $\sigma_Y^2=\mathbb{I}$. Columns 1 and 2 each produce one explicit minus sign from the third row. For column 3,

$$(\sigma_X\otimes\sigma_X)(\sigma_Z\otimes\sigma_Z)(\sigma_Y\otimes\sigma_Y)=(\sigma_X\sigma_Z\sigma_Y)\otimes(\sigma_X\sigma_Z\sigma_Y)=(-i\,\mathbb{I})\otimes(-i\,\mathbb{I})=-\,\mathbb{I}\otimes\mathbb{I}.$$

□

Property 1 says that Alice's three observables can be measured simultaneously, because commuting Hermitian operators share an eigenbasis. Property 2 says that on that common eigenbasis the product of her three outcomes is the eigenvalue of the product operator, which is $+1$. Property 3 says the same for Bob with the value -1. So conditions (a) and (b) of the game hold with certainty, whatever the state.

Condition (c), the agreement at the intersection, is where the shared state enters.

Lemma 6.2 (Transpose identity)**.** *Let $|\Phi_d\rangle=\frac{1}{\sqrt{d}}\sum_{k=0}^{d-1}|k\rangle|k\rangle$ and let M be any operator on $\mathbb{C}^d$. Then*

$$(M\otimes\mathbb{I})|\Phi_d\rangle=(\mathbb{I}\otimes M^{\mathsf{T}})|\Phi_d\rangle. \tag{25}$$

Proof. Write $M|k\rangle=\sum_l M_{lk}|l\rangle$. Then

$$(M\otimes\mathbb{I})|\Phi_d\rangle=\frac{1}{\sqrt{d}}\sum_{k,l}M_{lk}|l\rangle|k\rangle,\qquad(\mathbb{I}\otimes M^{\mathsf{T}})|\Phi_d\rangle=\frac{1}{\sqrt{d}}\sum_{l,k}(M^{\mathsf{T}})_{kl}|l\rangle|k\rangle,$$

and $(M^{\mathsf{T}})_{kl}=M_{lk}$. □

Now group Alice's two qubits into a single four-dimensional system A and Bob's into a single four-dimensional system B. The shared state is

$$\begin{aligned}|\Psi\rangle&=|\Phi^+\rangle_{A_1B_1}\otimes|\Phi^+\rangle_{A_2B_2}\\&=\frac{1}{2}\sum_{a,b\in\{0,1\}}|a\rangle_{A_1}|a\rangle_{B_1}|b\rangle_{A_2}|b\rangle_{B_2}=\frac{1}{2}\sum_{a,b\in\{0,1\}}|ab\rangle_A\,|ab\rangle_B,\end{aligned} \tag{26}$$

which is precisely $|\Phi_4\rangle$ in the basis $|ab\rangle$.

Proposition 6.3 (Perfect agreement at the intersection)**.** *For every cell (i,j), the operator O_{ij} of (24) has real entries, and*

$$(O_{ij}\otimes\mathbb{I}_4)\,|\Psi\rangle=(\mathbb{I}_4\otimes O_{ij})\,|\Psi\rangle. \tag{27}$$

Consequently $|\Psi\rangle$ is an eigenvector of $(O_{ij}\otimes\mathbb{I}_4)\,(\mathbb{I}_4\otimes O_{ij})$ with eigenvalue $+1$, so Alice's outcome and Bob's outcome at the cell (i,j) are equal with probability 1.

Proof. Each of σ_X, σ_Z and $\mathbb{I}$ is real, and $\sigma_Y \otimes \sigma_Y$ is real because the two factors of i multiply to -1. Every entry of (24) is therefore a real matrix, and being also Hermitian it is symmetric, so $O_{ij}^{\mathsf{T}} = O_{ij}$. Lemma 6.2 applied to $|\Phi_4\rangle = |\Psi\rangle$ gives the displayed identity. Applying it twice,

$$(O_{ij} \otimes \mathbb{I}_4)\,(\mathbb{I}_4 \otimes O_{ij})\,|\Psi\rangle = (O_{ij} \otimes \mathbb{I}_4)\,(O_{ij} \otimes \mathbb{I}_4)\,|\Psi\rangle = \left(O_{ij}^2 \otimes \mathbb{I}_4\right)|\Psi\rangle = |\Psi\rangle,$$

using $O_{ij}^2 = \mathbb{I}_4$. The product of the two outcomes is therefore $+1$ with certainty, and since each outcome lies in $\{+1, -1\}$ the two outcomes coincide. □

Conditions (a), (b) and (c) all hold with probability 1, so the quantum strategy wins every time.

Classical players separated in space are limited to 8/9. Players sharing two Bell states win with certainty. The two players never communicate, and the advantage comes entirely from correlations that no shared classical variable can reproduce.

Additional comment 6.2. A sign trap worth knowing about

A shorter-looking route to Proposition 6.3 writes $O_{ij} = P \otimes Q$ and factorises the correlation as

$$\langle\Phi^+|P \otimes P|\Phi^+\rangle \cdot \langle\Phi^+|Q \otimes Q|\Phi^+\rangle,$$

and then claims that $\langle\Phi^+|\sigma \otimes \sigma|\Phi^+\rangle = +1$ for every Pauli matrix σ. That claim is false. The correct values are

$$\langle\Phi^+|\sigma_X \otimes \sigma_X|\Phi^+\rangle = +1, \quad \langle\Phi^+|\sigma_Y \otimes \sigma_Y|\Phi^+\rangle = -1, \quad \langle\Phi^+|\sigma_Z \otimes \sigma_Z|\Phi^+\rangle = +1.$$

The sign at σ_Y comes from $\sigma_Y^{\mathsf{T}} = -\sigma_Y$ in Lemma 6.2. The conclusion survives only because the single cell of the square containing σ_Y is $\sigma_Y \otimes \sigma_Y$, where the two minus signs cancel. The proof given above avoids the issue entirely, since it uses $O_{ij}^{\mathsf{T}} = O_{ij}$ for the full two-qubit observable rather than factor by factor.

Additional comment 6.3. No information is transmitted

It is worth checking that the strategy does not smuggle in communication. Bob's measurement acts as $\mathbb{I}_4 \otimes (\cdot)$ on $|\Psi\rangle$, so the reduced state on Alice's side is unchanged by anything Bob does, and the distribution of Alice's three outcomes is uniform over the four triples with product $+1$ whatever column Bob was asked about. Alice therefore learns nothing about j, and symmetrically Bob learns nothing about i. Entanglement creates correlations that are stronger than classical ones without allowing a signal. Both statements are needed, and both are true.

Additional comment 6.4. What the square proves besides winning a game

The array (24) is a proof of the Kochen-Specker theorem in dimension 4. Suppose one tried to assign a definite value in $\{+1, -1\}$ to every observable in advance, consistently with the algebraic relations that commuting observables satisfy. Proposition 6.1 would force those values to satisfy (18), which we proved impossible. So the observables cannot all have predetermined values independent of which other compatible observables are measured alongside them. This is called contextuality, and the magic square proves it without reference to any particular state, which is stronger than the usual Bell inequality argument.

Additional comment 6.5. Relation to the Bell and CHSH inequalities

The more familiar Bell test, in the CHSH form, is a two-question game whose classical optimum is 0.75 and whose quantum optimum is $\cos^2(\pi/8) \approx 0.8536$. It separates the two theories statistically, so an experiment needs many rounds and careful statistics to distinguish 0.75 from 0.8536. The magic square is sharper. The classical bound is 8/9 and the quantum value is exactly 1, so a single

observed run in which the players win a query they are classically forced to lose is already meaningful. Both games have been realised experimentally with the outcomes predicted by quantum mechanics.